

FAHUD, Oman

WMO: 412620

Lat: 22.3475N

Lon: 56.4903E

Elev: 551

StdP: 14.41

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	52.1	54.5	18.7	14.3	84.9	22.6	17.3	86.5	26.5	72.7	24.6	72.4	5.9	270	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.8	115.9	70.4	114.0	70.5	112.2	70.4	77.4	96.5	76.7	96.9	76.0	96.6	8.1	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.4	126.7	84.6	71.9	120.4	84.6	70.6	115.0	84.8	41.5	97.7	40.7	96.3	39.9	96.4	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.5	22.4	19.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	84.9	67.8	71.3	77.9	86.4	95.1	98.2	98.9	96.2	92.1	86.3	77.3	71.0
	DBStd	11.69	3.79	4.94	5.43	4.98	3.74	3.31	3.62	4.34	3.43	3.65	3.69	3.77
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	26	15	5	0	0	0	0	0	0	0	0	0	6
	CDD50	12755	553	597	864	1091	1397	1447	1517	1431	1264	1124	820	650
	CDD65	7305	102	182	399	641	932	997	1052	966	814	659	370	191
	CDH74	113931	852	1894	4794	9411	15809	17523	18720	16582	13146	9627	3997	1576
	CDH80	77364	126	641	2406	5930	11463	13212	14268	12195	8990	5976	1773	384
Wind	WSAvg	7.9	6.7	9.0	9.3	7.7	8.9	9.1	8.8	9.5	8.6	5.7	5.2	6.7
Precipitation	PrecAvg	3.0	0.4	0.4	0.5	0.4	0.1	0.2	0.2	0.2	0.1	0.1	0.2	0.3
	PrecMax	5.4	0.7	1.3	3.7	2.2	0.8	1.4	0.9	0.7	0.8	0.2	0.9	1.2
	PrecMin	1.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.1	0.2	0.3	0.7	0.5	0.2	0.3	0.2	0.2	0.2	0.1	0.2	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.4	94.3	99.7	107.7	116.6	118.4	117.5	116.0	111.2	105.9	95.4	88.5
		MCWB	61.5	63.5	63.6	66.4	69.0	69.7	71.7	71.3	69.5	66.5	64.4	62.6
	2%	DB	83.1	90.0	96.3	104.8	112.9	116.1	115.6	113.6	108.7	103.4	93.3	85.9
		MCWB	61.5	62.3	63.6	65.9	68.1	69.4	71.4	71.7	69.7	66.5	64.2	63.0
	5%	DB	80.9	86.5	94.0	102.3	110.2	113.8	113.7	111.6	106.7	101.1	91.2	84.2
		MCWB	61.2	61.7	63.5	65.7	67.9	69.2	71.4	72.1	69.7	66.4	64.6	63.4
	10%	DB	78.9	83.8	91.5	100.0	108.0	111.6	111.7	109.5	104.4	98.7	88.8	82.1
		MCWB	60.2	61.4	63.3	64.8	67.6	69.0	71.5	72.2	69.7	65.8	64.3	62.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.4	68.3	73.5	75.7	77.4	78.8	78.5	77.9	76.3	75.6	72.4	68.8
		MCDB	74.3	78.9	81.8	86.9	90.2	91.7	100.1	101.4	94.4	89.0	82.4	76.2
	2%	WB	66.1	66.7	71.2	73.2	76.0	77.4	77.3	76.5	75.3	73.9	70.1	67.4
		MCDB	75.6	79.7	80.8	85.8	90.0	93.2	99.6	101.0	94.7	88.6	81.5	77.3
	5%	WB	64.3	65.5	69.6	71.5	74.5	76.5	76.5	75.8	74.5	72.3	68.6	66.1
		MCDB	76.4	79.7	81.4	86.4	91.5	94.0	99.1	99.6	93.4	88.6	81.9	78.0
	10%	WB	62.4	64.0	68.0	69.6	72.8	75.4	75.8	75.0	73.7	70.3	67.3	64.6
		MCDB	75.7	78.8	81.5	87.1	95.0	94.9	98.4	98.3	91.9	88.8	82.2	78.5
Mean Daily Temperature Range	5% DB	MDBR	23.6	25.6	25.3	26.0	26.7	27.2	25.8	25.4	25.3	26.2	24.1	23.3
		MCDBR	24.9	27.6	27.4	27.6	28.6	29.4	27.4	26.4	27.1	27.8	25.6	24.4
		MCWBR	10.8	11.1	10.2	9.0	9.3	8.8	7.6	5.7	7.1	9.6	10.7	10.4
	5% WB	MCDBR	21.6	23.5	20.6	22.1	23.5	25.3	25.1	24.8	24.4	23.1	21.7	22.5
		MCWBR	10.7	10.6	9.0	9.5	9.0	8.4	6.2	5.0	5.5	9.4	10.5	11.1
Clear-Sky Solar Irradiance	taub		0.417	0.466	0.527	0.584	0.622	0.709	0.795	0.760	0.636	0.490	0.435	0.412
	taud		2.082	1.926	1.774	1.685	1.616	1.495	1.408	1.455	1.623	1.896	2.050	2.091
	Ebn at Noon		265	259	249	238	227	207	190	196	219	249	257	261
	Edh at Noon		47	58	71	79	84	94	103	98	82	59	48	45
All-Sky Solar Radiation	RadAvg		1520	1768	2022	2215	2382	2331	2121	2132	2043	1880	1590	1440
	RadStd		51	56	96	108	48	83	84	52	31	48	27	48

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page